

Fixed effects and clustered inference

Slide route: Lecture10v2.tex: Fixed Effects Models through Final Remarks on Clustered SEs. Chapter 10.

Session 14 | Learn from changes within units

Outcomes: Compute a within transformation, interpret unit and time effects, state strict exogeneity, and explain clustering.

100-minute route

0–10	Retrieve what first differences remove.
10–30	Fixed-effects regression and within transformation.
30–45	Work the two-unit example.
45–55	Pause and contrast within/between variation.
55–75	Unit and time fixed effects; assumptions.
75–90	Serial correlation and clustered SEs.
90–100	Exit diagnosis and binary-outcome preview.

Worked example | Demeaning

Use with slides Lecture10v2 slides 38–49: Fixed Effects Regressions through the fixed-effects assumptions. Use the two-unit demeaning table with slides 38–43, then contrast it with unit/time effects on slides 44–49.

Use synthetic observations for two units over three periods:

$$X_A = (1, 2, 3), \quad Y_A = (12, 14, 16), \quad X_B = (2, 3, 4), \quad Y_B = (24, 26, 28).$$

The unit means are $(\bar{X}_A, \bar{Y}_A) = (2, 14)$ and $(\bar{X}_B, \bar{Y}_B) = (3, 26)$. After subtracting unit means, both units have

$$\tilde{X} = (-1, 0, 1), \quad \tilde{Y} = (-2, 0, 2).$$

Thus the within slope is

$$\hat{\beta}_{\text{FE}} = \frac{\sum_{it} \tilde{X}_{it} \tilde{Y}_{it}}{\sum_{it} \tilde{X}_{it}^2} = \frac{8}{4} = 2.$$

Contrast The line through the two unit means has slope $(26 - 14)/(3 - 2) = 12$. Between-unit differences and within-unit changes can tell very different stories. The large unit-level intercept difference explains this constructed contrast.

Assumptions and inference

Use with slides Lecture10v2 slides 45–54: accounting for unit effects, time effects, two-way fixed effects, strict exogeneity, serial correlation, clustered SEs, and final clustered-SE remarks. Use the school-clustering example during slides 50–54.

Start with

$$Y_{it} = \alpha_i + \lambda_t + \beta X_{it} + u_{it}.$$

Unit effects remove fixed differences; time effects absorb common time shocks. In a balanced panel, the two-way transformed variable is

$$\ddot{Y}_{it} = Y_{it} - \bar{Y}_i - \bar{Y}_t + \bar{Y}$$

and similarly for X . Use appropriate regression/projection machinery for an unbalanced panel rather than assuming this shortcut applies unchanged.

Strict exogeneity A familiar sufficient condition is $\mathbb{E}(u_{it} \mid X_{i1}, \dots, X_{iT}, \alpha_i) = 0$, with common time effects accounted for. Unobserved shocks that change later treatment choices can violate it. Fixed effects do not remove arbitrary reverse causality.

What cannot be identified A regressor constant within a unit is collinear with unit effects. A regressor identical across units within each period is absorbed by time effects. Ask which variation remains before interpreting a slope.

Clustering side example

There are 50 schools with 20 pupil observations each. Errors within a school may share a school-level shock. Treating 1,000 pupils as 1,000 independent pieces of information can overstate precision.

Say carefully Cluster-robust inference permits within-cluster dependence under its assumptions while relying on adequate independent cluster information. Fifty is not a universal adequacy threshold; few or unbalanced clusters can require special treatment.

Illustrative command For a genuine panel with identifiers `id` and time:

```
xtset id time
xtreg y x i.time, fe vce(cluster id)
```

Exit answers Fixed effects change the source of identifying variation. Clustered SEs address inference under dependence; they do not fix endogenous regressors. Adding time dummies does not absorb unit-specific changing confounders.

If short / preparation Reduce repeated diagrams while retaining the within calculation and clustering intuition. Read Chapter 11 and ask students why OLS might predict a probability above one.

Source: Lecture10v2.tex, fixed-effects and clustered-SE sections; synthetic examples.